MILITARY SPECIFICATION

TEST SET, GUIDED WEAPON AN/DSM-139A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the Guided Weapon Test Set AN/DSM-139A, herein referred to as the test set. The test set is for flightline use and perform data link operational readiness check (GO/NO-GO) of WALLEYE weapons defined in 6.1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-535	- Boxes, Shipping, Fiberboard

Military

MIL-P-116	- Preservation-Packaging, Methods of
MIL-E-5400	- Electronic Equipment; Aerospace and Guided Missiles Applicable Documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-T-5422	Testing, Environmental, Airborne Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life, for Aircraft Electronic Equipment, Format for
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

STANDARDS

Federal

FED STD 123	Marking for Shipment (Civil Agencies)

Military

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-167	Mechnical Vibration of Shipboard Equipment
MIL-STD-454	Electronic Equipment, General Requirements for
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

DL 910AS100	- Data List for Guided Weapon Test Set AN/DSM-139A

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturer's in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.2 Parts and materials.

3.2.1 General. The parts and materials specified in the drawings and related documents for use in the construction of this equipment were selected to ensure the fulfillment of all the requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change and the associated cost upon the contractor.

3.2.2 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965.

3.2.3 Supplemental requirements. In the event that the contractor proposes to alter the equipment delineated by this specification and applicable drawings, the supplemental requirements specified in 3.2.3.1 and 3.2.3.2 for the selection of parts and materials shall be applicable.

3.2.3.1 Basic conformance. Parts and materials selected for use in the equipment shall be as specified in MIL-T-28800. Parts shall be selected from those conforming to MIL-E-5400. If the parts listed therein will not perform adequately in the intended applications and environments specified herein see 3.2.2.

3.2.3.2 Interchangeability. All parts and materials shall meet the interchangeability requirements as specified in MIL-T-28800. The drawings conforming to DL 91OAS1OO shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies and parts.

3.3 Design and construction. The test set shall be manufactured in accordance with the drawings and subsidiary specifications conforming to DL 91OAS1OO, except that errors of omission or commission on the drawings may be incompatible with the requirements of this specification. In any case, it is a condition of final acceptance that the equipment manufactured shall meet all the design, construction, and workmanship requirements of this specification.

3.3.1 Reliability.

3.3.1.1 Operational stability. The test set shall operate as required, continuously or intermittently, for a period of at least 100 hours. During

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this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use.

3.3.1.2 Operating life. The test set shall have a minimum total operating of 1000 hours with minimal servicing and replacement of parts.

3.3.1.3 Reliability in mean-time-between-failures (MTBF). The test set shall have a specified mean [operating) time between failures of 100 hours when tested in accordance with 4.5.3.

3.3.2 Standard conditions. The following conditions shall be used to establish the normal performance characteristics under standard conditions for performing laboratory bench tests:

a.	Temperature	Room ambient (30 ± 10°C)
b.	Altitude	Normal ground (10,000 ft. maximum)
c.	Vibration .	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power Voltage	Fully charged battery pack (PN 178AS200) or external power source capable of producing 200 mA of current at 10.0 to 15.0 volts dc with a maximum ripple voltage of 1 mV rms.

3.3.2.1 Service conditions. The test set shall meet the operating test requirements of this specification under any environmental service-condition or combination of conditions specified in MIL-T-28800 for class 2 equipment, except as modified herein.

3.3.2.1.1 Temperature. The test set, with battery pack (PN 178AS200) installed, shall meet all operating test requirements of this specification when tested at the limited temperatures of O to 55°C. The test set with battery pack (PN 178AS200) installed shall meet all electrical requirements for a minimum of 30 minutes when tested at temperatures of O to -40°C under the following conditions:

a. The test set, less battery pack (PN 178AS200), shall be stabilized at required test temperature.

b. Battery pack (PN 178AS200) shall be charged at 60 ma for 16 hours at standard conditions.

c. Battery pack (PN 178AS200) shall be placed in test set immediately after charging and the tests initiated after the battery has been subjected to the test temperatures for a maximum of 15 minutes.

3.3.2.1.1.1 Battery pack. The battery pack shall meet the requirements of drawing 178AS200.

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3.3.2.1 .1.2 External power source. The test set shall be capable of unrestricted compliance with the MIL-T-28800 operational temperature requirements for class 2 equipment when powered from the external power source specified in 3.3.4 in lieu of battery pack PN 178AS200.

3.3.2.1.2 Drip-proof. The test set shall meet the test requirements of MIL-STD-109 for drip proof test except the equipment shall be inclined only 15 degrees. The equipment shall operate during and after exposure.

3.3.2.1.3 Humidity. The test set shall meet the test requirements of MIL-STD-81O, Method 507.1, Procedure I, including Method 507.1 paragraph entitled Summary, except the maximum temperature shall be 50°C.

3.3.2.1.4 Salt spray. The test set shall meet the test requirements of MIL-STD-81O, Method 509.1, Procedure 1, including Method 509.1 paragraph entitled Summary.

3.3.2.1.5 Explosive atmosphere. The test set shall meet the test requirements of MIL-STD-81O Method 511.1, Procedure I, including Method 511.1 paragraph entitled, Summary, except the maximum temperature shall be 55°C and the maximum altitude shall be 10,000 feet (3,003 meters).

3.3.2.1.6 Vibration. The test set shall meet the test requirements of MIL-STD-28800 for Class 2 equipment.

3.3.2.1.7 Shock. The test set shall meet the test requirements of MIL-STD-810, Method 516.1, Procedure I, including Method 516.1 paragraph entitled Summary. The equipment shall not be operating during the test.

3.3.3 Warm-up time. The electronics assembly shall perform its required functions within the time periods specified in 3.3.3.1 through 3.3.3.4.

3.3.3.1 Frequency sweep. The frequency shall be swept across the F_{oc} ± 2.25 MHz band within 30 seconds after S1 or S2 is depressed. All requirements of the swept frequency specified in 3.5. 1.2.1 shall be met.

3.3.3.2 Detection of RF signal. Lamp, DS-1, shall provide an indication within five seconds after appropriate signal is applied and S1 is depressed.

3.3.3.3 Sweep and detector circuit self test. DS-1 shall indicate sweep circuit and detector circuit (excluding detector diode) are functioning within 30 seconds after S2 is depressed.

3.3.3.4 Battery self test. DS-2 shall indicate battery condition within five seconds after S1 or S2 is depressed.

3.3.4 Primary input power requirements. The power required shall be one of the following types and shall be not greater than the specified amounts:

a. A fully charged battery pack which meets the requirements of drawing 178AS200.

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b. A 220 ± 10 volts, 50 ± 10 Hz power source (for operation of battery charging circuit also).

3.3.5 Electromagnetic interference suppression. The test set is intended to function properly and survive without degradation or damage to interior component parts when subjected to electromagnetic environments existing on flight decks of aircraft carriers. In addition, electromagnetic radiation and susceptibility requirements of MIL-STD-461 shall be met except as specified in 3.3.5.1 and 3.3.5.2.

3.3.5.1 CE06. Narrowband conducted emissions to the video receiving antenna may exceed the specification limits by not more than 4 db at 3 F_{oc} and 3 db at 5 Foe. F_{oc} is the RF frequency of the voltage controlled oscillator (VCO) which generates the command signal when either S1 or S2 is depressed.

3.3.5.2 RE02. The test set shall meet all the requirements for RE02 steady state. Extension of RE02 limits for switching transients due to activation of S1 and S2 shall be 12 db at 40 to 60 MHz for S2 and 10 db at 170 to 190 MHz for S1 when the plain shroud is attached; and, shall be 6 db at 260 to 280 MHz for S2 and 12 db at 40 to 60 MHz for S1 when EMCON shroud is attached.

3.3.6 Test set case dimensions. The dimensions of the test set case shall be as follows:

a. Length - 22-1/2 inches (57.2 cm)
b. Width - 19 inches (48.3 cm)
c. Depth - 13-1/2 inches (34.3 cm)

3.3.7 Weight. The total weight of the test set, including cables shall be not greater than 40 pounds (18.144 kgs).

3.3.8 Color and finish. The color and finish of the test set shall be in accordance with DL 910AS100.

3.3.9 Identification plate and item markings. Nomenclature and identification marking shall be in accordance with MIL-STD-130 and as specified in DL 91OAS1OO.

3.4 Performance. The performance of the test set shall be as specified herein in addition to the performance requirements specified in DL 91OAS1OO. Unless otherwise specified herein, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations will be specified.

3.4.1 Test set functions. The test set shall perform the functions specified in 3.4.1.1 through 3.4.1.3.

3.4.1.1 WPN TEST switch. The WPN TEST switch S1 shall energize all active circuits In the test set to check battery condition, interrogate weapon and receive weapon RF channel signal.

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3.4.1.2 Weapon test mode. In the weapon test mode, the test set shall radiate a modulated RF carrier and shall sweep the carrier through the frequency band of the command receivers of the WALLEYE weapons. Indicator lamp DS1 (green) shall flash on at approximately one second intervals, interrupted for 5 to 18 second intervals in each 20 second sweep period.

3.4.1.3 SELF TEST switch. The SELF TEST switch S2 shall energize circuitry to check the condition of the battery pack and all active circuits in the test set in the video detector (excluding the detector diode), the two subcarrier oscillators and the voltage controlled oscillator. A yellow SELF TEST lamp (DS2) shall flash on at approximately two second intervals to indicate the battery pack is in operating condition.

3.5 Detail requirements.

3.5.1 Test set subassemblies. The test set shall consist of the following subassemblies (listed in DL 1098AS1OO) and shall be manufactured in accordance with the indicated drawings and all-documents listed thereon, The test set shall meet the requirements stated on the indicated drawings and all documents listed thereon:

a.	Complete Test Set	91OAS1OO
b.	Transit Case	91OAS2OO
c.	Electronics Assembly	91OAS3OO
d.	Shroud Assembly, Plain	887AS400
e.	Shroud Assembly, EMCON	887AS500

3.5.1.1 Detail functions. The electronics assembly shall simultaneously radiate a frequency modulated command signal and shall detect a radiated carrier signal which may be unmodulated or frequency modulated with a video signal . The electronics assembly shall perform its radiation and detection functions when neither shroud is attached or when either shroud is attached. The plain shroud, when attached to the electronics assembly, shall provide protection from high intensity electromagnetic fields for the detector system in the electronics assembly; and, shall absorb RF energy from the antenna on the system under test when the shroud is placed over this antenna. The EMCON shroud shall perform the same functions as the plain shroud except it shall provide additional protection for the detector system and shall minimize radiation from the antenna on the system under test. The electronics assembly shall have self test and battery charging capability.

3.5.1.2 Electronics assembly. The electronics assembly shall contain the radiate and receive antennas, command signal generation subsystem, detector subsystem, 1? volt battery pack, battery test and self test subsystem, battery charging subsystem and the control and indicator subsystem.

3.5.1 .2.1 Command subsystem. When weapon test S1 switch is depressed, the electronics assembly shall generate and radiate a frequency modulated RF command signal which has the following characteristics:

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3.5.1.2.1.1 Frequency (command subsystem). Swept over a range of F_{oc} -2.25 MHz or lower to F_o +2.25 MHz or higher. F_{oc} is a frequency which will be specified by the procuring activity.

3.5.1.2.1.2 Sweep characteristics. The total sweep period including retrace and dead time between end of retrace and active sweep shall be 20 ± 3 seconds. The sweep rate shall be not greater than 2.5 MHz per second in the F_{oc} ± 2.25 MHz band. The slope of the frequency vs. time characteristic shall be positive for the active portion of the sweep.

3.5.1.2.1.3 Modulation. The carrier shall be modulated by 2300 and 5400 Hz subcarrier oscillators.

3.5.1.2.1.4 Carrier deviation. The peak-to-peak carrier deviation shall be a minimum of 15 KHz for the 2300 Hz frequency and 25 KHz for the 5400 Hz frequency with the peak-to-peak voltages at TP-9 and TP-10 set for 300 and 700 mV peak-to-peak respectively.

3.5.1.2.1.5 Radiated power. Radiated power shall be -40 ± 8 dbm.

3.5.1.2.1.6 Full sweep capability. The sweep shall complete its excursion across the F_{oc} ± 2.25 MHz band when S1 and S2 are released while the frequency is within this band, and shall not repeat until either S1 or S2 is again depressed.

3.5.1.2.2 Detector subsystem. When S1 is depressed, the electronics assembly shall detect and indicate the presence of an unmodulated or frequency modulated RF signal as specified in 3 .5.1.2.2.1 through 3.5.1.2.2.4.

3.5.1.2.2.1 Frequency (detector subsystem). Any (and all) frequency(ies) in a band of F_{ov} ± 20 MHz. F_{oc} is a frequency which will be specified in the contract (see 6.2.1).

3.5.1 .2.2.2 Power level (detector subsystem). -1 to +10 dbm applied directly to the detector input at P1.

3.5.1.2.2.3 Sensitivity adjustment (detector subsystem). Adjustment of variable resistor, R6, shall provide a threshold sensitivity adjustment over a minimum range of -1 to +3 dbm when the signal is applied directly to the input of the detector at PI.

3.5.1.2.2.4 Detection indication (detector subsystem). An indication that the electronics assembly has detected a signal, as defined above, shall be provided by indicator lamp, DS 1. The lamp shall start flashing within 5 seconds after application of the RF signal and shall continue with a period of +1, -.5 second and duty cycle of 10 to 25 percent.

3.5.1 .2.3 Self test subsystem. The electronics assembly shall contain self test capability which tests the sweep circuit, the detector circuit and battery condition as specified in 3.5.1.2.3.1 through 3.5.1.2.3.3.

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3.5.1.2.3.1 Sweep circuit and detector circuit. When S2 is depressed, the sweep voltage shall be generated and applied to the input of the detector circuit, excluding the detector diode. The sweep voltage shall activate the detector circuit and cause indicator lamp DS 1 to flash with a period of 1 ± .5 second and duty cycle of 10 to 25 percent during a portion of the active sweep. The number of flashes per sweep period shall be a function of the adjustment of R6, the battery voltage, and sweep rate and sensitivity variation. The number of flashes per sweep shall be as specified in table I.

TABLE 1. Number of flashes per sweep.

R6 Adjustment	Battery Voltage		
	10 Volt	12.5 Volt	15 Volt
Fully Counter Clockwise	13-15	14-16	14-17
At Mid Range	8-10	10-12	11-14
Fully Clockwise	4-6	7-9	8-11

3.5.1.2.3.2 Battery test. When S1 and/or S2 is (are) depressed, the DS-2 indicator lamp shall lash with a period of 2 ± 1 seconds and duty cycle of 10 to 25 percent when the battery voltage is in the range of 10.15 ± .15 volts dc to 15 volts dc. It shall not flash when the battery voltage is 10 volts dc or less.

3.5.1.2.3.3 Battery condition indicator threshold adjustment. Adjustment of variable resistor, R12, shall provide a battery condition indicator threshold range of at least 10 to 10.3 volts.

3.5.1.2.4 Battery charging subsystem. The electronics assembly shall contain battery charging capability. A 220 ± 10 volt, 50 ± 10 Hz power source shall be connected to the test set at connector J101 via the battery charging cable. The test set shall provide a 7.7 ± .80 volt peak half sine wave across a 51 ohm resistor connected across J2 pins 7 and 4 or pins 14 and 11. Pins 14 and 7 shall be positive with respect to pins 11 and 4.

3.5.1.3 Plain shroud. The plain shroud shall be easily attachable/ detachable to/from the electronics assembly. It shall be manufactured in accordance with drawing 887AS400. The shroud shall provide protection for the detector subsystem in the electronics assembly from high level electromagnetic fields and shall absorb and attenuate RF energy radiated from the antenna on the unit under test. The shroud shall provide at least 4 db attenuation of the RF signal which is radiated from the weapon during a weapon test.

3.5.1.4 EMCON shroud. The EMCON shroud shall be easily attachable! detachable to/from the electronics assembly. It shall be manufactured in accordance with drawing 887AS500. The shroud shall provide at least 48 db attenuation of the RF signal radiated from the weapon during a weapon test. The EMCON shroud shall be similar to the plain shroud except it shall have the following additional features:

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3.5.1 .4.1 Door and door operating mechanism. The EMCON shroud shall have a hinged door at the forward end and an opening - closing - locking mechanism suitable for opening or closing the door and locking it in both the closed and open positions.

3.5.1.4.2 RFI finger stock. The shroud shall have fingerstock which, when the door is closed, shall contact a mating surface on the antenna of the system under test; the fingerstock shall be arranged for 360° contact with this surface.

3.5.1.4.3 Strap assembly. The EMCON shroud shall contain a strap and handle assembly (drawing 887AS520 of DL 910AS100) for holding the shroud in place on the unit under test and applying pressure to the shroud/antenna interface to effect an EMI seal.

3.5.1.4.4 Electrical performance. When the shroud is attached to the electronics assembly and strapped onto the unit under test, the shroud shall reduce radiation from the tested antenna by at least 48 db at any (and all) point(s) in space.

3.5.2 Weight. Heights shall be not greater than as specified in table II.

TABLE II. Equipment weights.

Assembly	Drawing Number	Weight
Electronic Assembly	910AS300	6 lb. (2.72 kilograms)
Plain Shrould	887AS400	2.5 lb. (1.13 kilograms)
EMCON Shrould	887AS500	5 lb. (2.27 kilograms)
Transit Case	910AS200	25 lb. (11.35 kilograms)
Battery Pack	178AS200	17 oz. (0.482 kilograms)

3.5.3 Content. The test set shall contain the major assemblies specified in 3.5.3.1 through 3.5.3.4.

3.5.3.1 Electronics assembly (910AS300). The electronics assembly shall contain the subassemblies specified in 3.5.3.1.1 through 3.5.3.4.

3.5.3.1.1 Chassis assembly (910AS302). The chassis assembly shall contain the printed circuit board, subcarrier oscillators, voltage controlled oscillator, RF detector, battery pack, battery charger, 40 db attenuator, control switches, indicator lamps and associated cables.

3.5.3.1.2 Case assembly (887AS340). The case assembly shall contain the main case for the electronics assembly, the radiate and receive antennas, RF cables, radome and RF absorber material.

3.5.3.2 Plain shroud assembly (887AS400). The plain shroud assembly shall contain a housing, radome, RF absorber and antenna hood.

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3.5.3.3 EMCON shroud assembly (887AS500). The EMCON shroud assembly shall contain a housing, radome, RF absorber, antenna hood, retainer strap assembly, and a RFI seal subassembly which shall consist of RFI fingerstock and retainer, a hinged door and a door operating mechanism.

3.5.3.4 Transit case (91OAS2OO). The transit case shall consist of the case assembly and cushioning material.

3.5.4 Controls. Two pushbutton control switches shall be located on the front panel. The "WPN TEST" pushbutton, when depressed, enables the command, battery self test and RF detection subsystems. The "SELF TEST" pushbutton, when depressed, shall enable the command and battery self test subsystems and shall apply a sweep voltage to activate the RF detector circuit. When both pushbuttons are depressed, the battery self test subsystem shall be enabled.

3.5.5 Material. Materials shall conform to DL 91OAS1OO.

3.5.6 Workmanship Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the Inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspections (4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the conditions specified in 3.3,4.

4.3.1 Performance characteristics.

4.3.1.1 Before testing. The test set shall satisfy the requirements for interference control (see 3.3.5) and warm-up time (see 3.3.3).

4.3.1.2 During inspection acceptance testing. The test set shall satisfy the requirements of Section 3.

4.4 First article inspection. First article tests shall be made on a test set representative of the production test sets to be supplied under the con-

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tract. First article test performance shall be the responsibility of the contractor and shall be performed in accordance with approved test procedures (see 4.8). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the contracting officer.

4.4.1 First article test data. The contractor shall make available to the procuring activity all data collected in conducting these tests (see 6.2.2). On first production contracts, this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.4.2 Scope of first article tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the test set meets all the requirements' of this-specification and the contract. These shall include environmental and interference tests in accordance with MIL-T-5422 as specified in MIL-T-28800. Test methods shall be in accordance with MIL-STD-81O except as modified by this specification.

4.4.3 First article approval. Approval of the first article shall be by the procuring activity upon the contractor meeting the requirements of all tests and making available the first article test report. No production test sets shall be delivered prior to the approval of the first article. Prefabrication of production equipment prior to the approval of the first article is at the contractor's own risk. The approved first article sample shall be retained by the contractor for his use in the fabrication and testing of test sets to be submitted for acceptance. The first article sample shall not be considered as one of the deliverable test sets under the contract.

4.4.4 Production test sets. Test sets supplied under the contract shall, in all respects including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each test set shall be capable of successfully completing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection of a test set and of a test set already accepted by the Government. The contractor shall make necessary corrections as approved by the procuring activity.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing shall be to the approved quality conformance test procedures under the supervision of the Government inspector. The contractor shall make available test reports showing quantitative results for all quality conformance tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Quality conformance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Tests shall consist of the following:

a. Individual tests (see 4.5.1)

b. Sampling tests (see 4.5.2) or reliability assurance tests (see 4.5.3)

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c. Special tests (see 4.5.4)

d. Life tests (see 4.5.6)

4.5.1 Individual tests. Each test set made available for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each test set accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.l.2)

c. Manufacturing run-in test (see 4.5.1.3)

4.5.1.1 Examination of product. Each test set shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each test set shall be operated long enough to permit the temperature to stabilize and to check sufficient characteristics and record adequate data to assure required test set operation.

4.5.1.3 Manufacturing run-in test. Each test set shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made using controls that are accessible to the operator during normal use of the equipment.

a.	Temperature	Room ambient
b.	Humidity	Room ambient
c.	Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3 g's.

The test set shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the six-hour period of operation. Where feasible, the test set shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of required performance is not necessarily required during the vibration period. The direction of vibration shall be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the test set in two directions, the vertical shall be used. During the six-hour period of operation following the 10-minute vibration period, the test set shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, the test set shall be repaired and the test started over, except that the 13-minute vibration period need not be repeated when it is established that the failure was not a result of vibration. Should failures occur repeatedly, corrective action shall be taken to eliminate this defect from future test

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sets. The six-hour period specified above may be made up of several shorter periods to conform to normal working hours.

4.5.1.4 Temperature tests. Temperature tests shall be performed on each electronic assembly (910AS300) at -40 and +55°C following the manufacturing run in tests as follows:

a. Soak periods shall be one hour at -40 or at +55°C as applicable with test set unenergized.

b. A dc power supply capable of supplying 200 ma, 9.9 to 14.5 volts with 1 mv maximum ripple shall be substituted for the battery pack.

c. The battery test subsystem requirements (see 3.5.1.2.4) shall be verified with supply voltage at 9.95 ± .05 volts and at 10.35 + .05 volts.

d. The command subsystem requirements (see 3.5.1.2.1) and detector subsystem requirements (see 3.5.1.2.2) shall be verified with supply voltage at 10.35 ± .05 and at 14 ± .05 volts.

e. The self test subsystem (see 3.5.1.2.3) shall be verified at 10.35 ± .05 volts.

f. Following satisfactory conclusion of the temperature tests, the operationial test shall again be performed at room ambient conditions using battery pack 178AS200.

4.5.2 Sampling tests. Test sets selected for sampling tests shall first have passed the individual tests. (Sampling tests shall not be conducted unless reliability assurance tests of 4.5.3 are deleted by contractual action.) Equipments will be selected for sampling tests by the Government inspector in accordance with table III.

TABLE III. Sampling test sample selection.

Quantity of equipments offered for acceptance	Quantity to be selected for sampling tests
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 290 of fraction thereof.

4.5.2.1 Scope of tests. As a minimum, each test set selected for sampling tests shall be subjected to the following:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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b. Operational test at certain environmental conditions. The conditions may vary for each test set tested and should be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future test sets.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with the requirements of MIL-STD-781. Tests shall be conducted as required for both the qualification phase and the production acceptance (sampling) phase.

4.5.3.1 Reliability qualification phase. Prior to acceptance of test sets under the contract or purchase order, a contractually specified quantity of test sets (or a minimum of three if not specified) shall be subjected to the combined reliability qualification requirements for unsheltered shipboard equipment (category 3.B) of MIL-STD-781, using Test Plan IC therein (θ_1 = 200).

4.5.3.2 Reliability acceptance phase. Production lot samples of the test sets shall be subjected to the combined production reliability acceptance requirements for unsheltered shipboard equipment (category 3.B) of MIL-STD-781, using Test Plan VC therein (r_1 = 100).

4 .5.3.2.1 Acceptance phase sample size. The entire contract quantity (except for those test sets used in the qualification phase) shall be considered one lot. The reliability acceptance sample size shall be 10 percent of the contract quantity, but not less than three units. The test program shall start the first month after the qualification phase has been completed. The samples shall be tested until an accept/reject decision has been reached. In the meantime, other equipments manufactured may be shipped in place, with approval from the contracting officer. The test results shall be summarized monthly and made available for the procuring activity (see 6.2.2). The procuring activity reserves the right to stop the acceptance of test sets anytime after one more reject decisions have been reached, pending a review of the contractor's efforts to improve the test set, parts, quality control; in order that the entire contract quantity will result in an accept decision.

4.5.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be made available and accepted by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2}.

4 .5.3.3.1 Temperature extremes. The temperature extremes shall be limited to -40°C to +55°C storage, and -28°C to +55°C operating.

4.5.3.3.2 Voltage cycling. Voltage cycling in accordance with MIL-STD-781 is not required.

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4.5.4 Special tests. Special tests shall be conducted on a quantity of test sets as specified in the contract (see 6.2.1) for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure quality control. The test sets selected for special tests may be from test sets previously subjected to the sampling or reliability assurance tests.

4.5.4.1 Special test schedule. Selection of test sets for special tests shall be made as follows:

a. On early equipments after an engineering or material change.

b. Whenever failure reports or other evidences indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously approved for the first article tests may be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available it to the procuring activity for approval prior to conducting the tests (see 6.2.2).

4.5.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or due to design defect.

c. Make available to the procuring activity for acceptance the proposed corrective action intended to reduce the possibility of the same failure occurring in future tests (see 6.2.2).

d. Where practical, include a test (in the individual test) to check all test sets for this requirement until assurance is forthcoming that the defect has been corrected.

4.5.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.6.1. The life test sample will be selected by the Government inspector in accordance with table IV. (Equipments which have successfully completed the sampling tests, reliability tests, or special tests may be selected for life tests.)

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TABLE IV. Life test sample selection. 1/

Quantity of equipments offered for acceptance	Quantity to be selected for life test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof.

1/ When reliability tests are conducted, the life test may be omitted if, during the reliability tests, quantity of equipments equal to or more than that listed above receive at least 300 hours each of test time.

4.5.6.1 Life test conditions. The life test shall be conducted under the following simulated service conditions:

a.	Temperature	Room ambient (30°C ± 10°C)
b.	Altitude	Normal ground (O - 5000 ft.) (0 - 1600 meters)
c.	Humidity	Room ambient up to 90 percent relative humidity
d.	Input voltage	Nominal
e.	Vibration	None

4.5.6.2 Life test period. The life test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the test set temperature to stabilize. Periodically, the test set shall be turned on and off several times, and put through its various phases of operation.

4.5.6.3 Performance check. At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check shall be proposed by the contractor and shall be made available to the procuring activity (see 6.2.2).

4.5.6.4 Failure report. In the event of a failure, the Government inspector shall be notified immediately. A report shall be made available to the procuring activity upon completion of the test (see 6.2.2). In this report, the contractor shall propose suitable and adequate design or material corrections for all failures. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Presubmission testing. No item, part, or complete equipment shall be delivered by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

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4.7 Environmental tests. The test set shall operate as required when subjected to the tests specified in 4.7.1 through 4.7.7.

4.7.1 Temperature - altitude. The test set shall be tested in accordance with MIL-STD-81O Method 504.1, Procedure 1, including Method 504.1 paragraph entitled Summary, except the following operating conditions shall be used in place of table 504.1-11. Adjustment of video controls shall be allowed.

a. Step la shall be omitted.

b. Step lb shall be at a temperature of -62°C and altitude of 50,000 feet (16,000 meters).

c. Step 2 shall be at a temperature of -40°C and altitude at standard ambient.

d. Step 3 shall be at a temperature of -40°C and altitude of 10,000 feet (3,000 meters).

e. Step 4 shall be omitted.

f. Step 5 shall be at a temperature of 85°C and standard ambient altitude.

g. Step 6 shall be at a temperature of 55°C and standard ambient altitude.

h. Step 7 shall be at a temperature of 71°C and altitude of 10,000 feet (3,000 meters) with an equipment ON-time of 20 minutes.

i. Step 8 thru 12 shall be omitted.

j. Step 13 shall be at a temperature of 88°C and altitude of 50,000 feet (16,000 meters). The equipment shall be non-operating, and shall remain in the chamber for 4 hours.

k. Step 14 shall be at a temperature of 25°C and standard ambient altitude.

l. Step 15 shall be at standard ambient conditions.

4.7.2 Drip-proof. The test set shall be drip-proof tested in accordance with MIL-STD-108 except the equipment shall be inclined only 15 degrees. The equipment shall be operated during and after exposure.

4.7.3 Humidity. The test set shall be humidity tested in accordance with MIL-STD-81O Method 507.1, Procedure 1, including Method 507.1 paragraph entitled Summary, except the maximum temperature shall be 50°C.

4,7.4 Salt spray. The test set shall be salt spray tested in accordance with MIL-STD-81O Method 509.1, Procedure 1, Including Method 509.1 paragraph entitled Summary.

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4.7.5 Explosive-atmosphere. The test set shall be explosive tested in accordance with MIL-STD-810, Method 511.1, Procedure 1, Including Method 511.1 paragraph entitled Summary, except the maximum temperature and altitude shall be 55°C and 10,000 feet (3,000 metters), respectively.

4.7.6 Vibration. The test set shall be subjected to vibration tests in accordance with MIL-STD-167, Type 1.

4.7.7 Shock. The test set shall be shock tested in accordance with MIL-STD-810, Method 516.1, Procedure 1, including Method 516.1 paragraph entitled Summary, except the equipment shall not be operated during the test.

4.8 Methods of examination and test. The procedures used for conducting first article tests and quality conformance tests shall be prepared by the contractor and made available to the procuring activity for review and acceptance (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract (see 5.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

5. PACKAGING

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as follows:

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order (see 6.2.1), preservation-packaging shall be in accordance with MIL-P-116, Method III.

5.3 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), packing shall be level A. The packaged test sets shall be packed in fiberboard containers conforming to PPP-B-535, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-635.

5.3.1 Unitized loads. Unitized loads, commensurate with the level of packing specified herein or in the contract, shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent possible. Exterior of container shall be marked as follows:

THIS TEST SET CONTAINS NICKEL-CADMIUM BATTERIES.
DO NOT STORE BELOW -40°F OR ABOVE +131°F
UNLESS BATTERIES ARE REMOVED.

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5.4 Marking. In addition to any special marking specified in the contract (see 6.2.1), each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129. When specified in the contract (see 5.2.1), the marking of domestic shipments shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The intended use of the Guided Weapon Test Set AN/DSM-139A is to perform a GO/NO-GO confidence check on the data link subsystem in data link modified WALLEYE weapons after the weapon has been installed on the carriage aircraft. The test set exercises both the command and video portions of the weapon data link subsystem.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Frequency (detector subsystem) (see 3.5.1.2.2.1)

c. Selection of applicable levels of packaging and packing (see 5.2, 5.3).

d. When a first article is required for inspection and approval (see 4.4, 6.3).

e. When test procedures are to be modified or expanded (see 4.8).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.
4.4.1	Report, First Article Inspection	DI-T-4902
4.5, 4.5.6.4.1	Reports, Production Inspection	DI-T-4904
4.5.3.2.1, 4.5.3.3	Procedures, Reliability Tests	DI-R-7035
4.5.4.2, 4.8, 4.5.6.3	Procedures, Production/Acceptance Inspection	DI-T-4903

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Paragraph no.	Data requirement title	Applicable DID no.
4.5.5c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
4.8	Procedure, First Article Inspection	DI-T-4901
4.5.6.4	Report, Failure	UDI-R-21141A

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L. , Vol. II AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Fores Center or as directed by the contracting officer.

6.3 First Article. When a first article inspection is required the item will be tested and should be a first article sample. The first article should be one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and approval of the documents' first article.

Preparing Activity:
Navy - AS

(Project 4935-N050)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-85396(AS)	Test Set, Guided Weapon AN/DSM-139A

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* - Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c MAILING ADDRESS *(Street City State ZIP Code)* - Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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